

Muhammad Rahim

Full-Stack Developer | AI/ML & Data Science | BS Computer Science
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LinkedIn: <https://linkedin.com/in/muhammadrahimusman> | **GitHub:** <https://github.com/Rebelhere> | **Portfolio:** <https://muhammadrahim.vercel.app>

Summary

Versatile Computer Science undergraduate at FAST-NUCES (CGPA 3.41, Dean’s List) with hands-on experience across full-stack web development, AI/ML systems, agentic LLM workflows, and data engineering. Proven track record shipping production-ready React.js + FastAPI platforms, integrating LLMs with real-time async backends, building RAG-based retrieval systems, and applying NLP preprocessing pipelines. Proficient in model fine-tuning (PEFT/QLoRA), LangChain/LangGraph orchestration, prompt engineering, CI/CD pipelines, and Docker-based deployments. Seeking internship opportunities at the intersection of software engineering and intelligent systems.

Education

FAST-NUCES
BS Computer Science
CGPA: 3.41/4.0 • Dean’s List — 3rd Semester 2024 & 5th Semester 2025 (Top 10% of cohort)
Relevant Coursework: Artificial Intelligence, Data Science, Natural Language Processing, Probability & Statistics, Data Structures & Algorithms, Database Systems, Web Technologies, Software Engineering

Lahore, Pakistan
Expected June 2027

Projects

ClankerFUED — AI-Integrated Multiplayer Game — *Python, LLM, Agentic Workflow, Async, Unity*

- Architected an agentic LLM backend enabling real-time, context-aware NPC dialogue for 2+ concurrent players — implementing memory handling, prompt engineering, and latency reduction via non-blocking async I/O to maintain coherent multi-turn game-state interactions.

Quran & Hadith Platform — RAG-Augmented Semantic Search — *FastAPI, React.js, SQL, NLP, RAG*

- Built a retrieval-augmented search system over 6,000+ structured Islamic text records, applying NLP preprocessing (tokenization, stop-word filtering, deduplication) to improve retrieval relevance ~30% over baseline keyword matching.
- Engineered FastAPI backend with SQLAlchemy ORM achieving sub-200ms query response; retrieval pipeline mirrors core RAG patterns of document ingestion, chunking, and context-augmented response generation.

ShopVERSE — Full-Stack Multi-Vendor E-Commerce Platform — *React.js, FastAPI, SQL, Docker*

- Built a modular React.js frontend with reusable components, state management, and mobile-first responsive layouts; implemented 15+ FastAPI REST endpoints with SQLAlchemy ORM, input validation, and a normalized multi-entity SQL schema for 5+ merchants.
- Designed dynamic filtering, search, and brand-management UI through Figma prototyping — reducing navigation depth by 40% over initial wireframes; structured for Docker-based containerized deployment.

Personal Portfolio — muhammadrahim.vercel.app — *React.js, CSS Animations, Vercel CI/CD*

- Designed and built a fully responsive portfolio with scroll-triggered CSS animations and mobile-first layout; deployed on Vercel with a CI/CD pipeline cutting deployment to under 2 minutes per push.

FAST-NUCES
Teaching Assistant — Database Management Systems
Guiding 40+ students through SQL query design, normalization, indexing, and schema modeling — reinforcing structured data engineering skills applicable to AI/ML pipelines, RAG knowledge-base construction, and backend systems.

Lahore, Pakistan
January 2026 – June 2026 (Ongoing)

FAST-NUCES
Teaching Assistant — Data Structures & Algorithms
Designed supplementary exercises for data structures and algorithms modules, improving average assessment scores and strengthening students’ algorithmic foundations essential for ML system and backend architecture design.

Lahore, Pakistan
August – December 2025

Technical Skills

AI / ML: Machine Learning, NLP, LLM Integration, Model Fine-Tuning, PEFT, QLoRA, RAG-Based Frameworks, Feature Engineering
Agentic & LLM Tooling: Agent Orchestration, Agentic Workflows, MCP Integrations, LangChain, LangGraph, Prompt Engineering, Chatbot Optimization, Memory Handling, Latency Reduction
Frontend: React.js, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Bootstrap, Responsive Design, Figma
Backend & APIs: FastAPI, Django, Python, RESTful APIs, Async Programming, WebSockets, Node.js
Data & Databases: NumPy, Pandas, SQLAlchemy, EDA, Query Optimization | MS SQL Server, MongoDB, SQLite, MySQL
DevOps & Tools: CI/CD Pipelines, Docker, Kubernetes, Git, GitHub, Postman, Linux
Languages: Python, SQL, JavaScript, C++